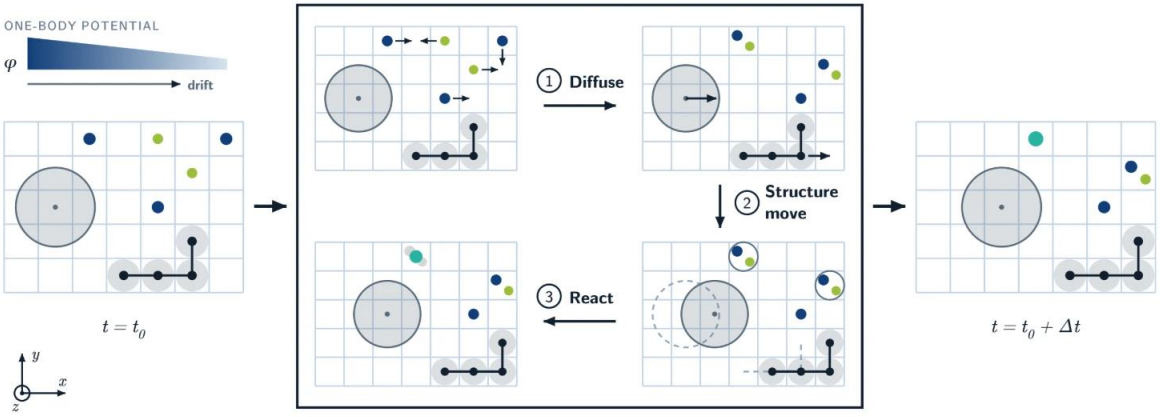
Lattice RDDME scheme with finite-size particles

a

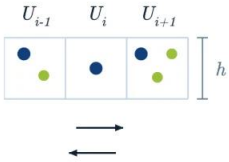
One timestep



b

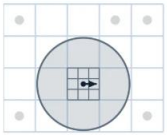
Moves within one timestep

1 Diffuse



$$p_{i\to i\pm1} = \frac{D}{h^2}\, B(U_{i\pm1} - U_i)$$

2 Structure move



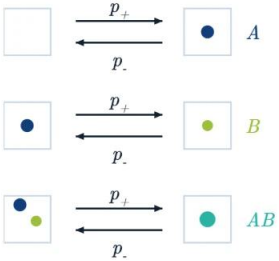
center of mass on a lattice site

crowding from surrounding species

$$p_{i\to i\pm\delta} = \frac{D}{\delta^2}\, f\, B(U_{i\pm\delta} - U_i)$$

$$f = \left[1 + \sum_s c_s \eta_s\right]^{-1}$$

3 React

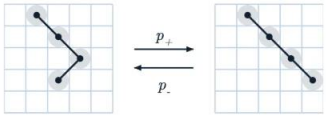


$$p_\pm = p_\pm^{(0)} \exp(U_f - U_r)$$

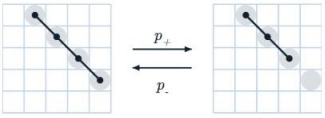
c

Constraints on structure moves

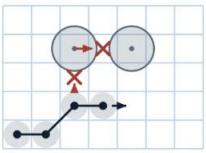
Chain conformation



Chain chemistry



Hard-sphere exclusion



rejected: target site overlaps another structure

accepted: target site has no structure